A. Increasing and Decreasing

time limit per test 1 second
memory limit per test 256 megabytes
input standard input
output standard output

Rating: 800 | Tags: constructive algorithms, greedy, implementation, math |
https://codeforces.com/problemset/problem/1864/A

You are given three integers x, y, and n.

Your task is to construct an array a consisting of n integers which satisfies the following conditions:

1. $a_1=x$, $a_n=y$;
2. a is **strictly** increasing (i.e. a_1 lt a_2 lt ... lt a_n);
3. if we denote $b_i=a_{i+1}-a_i$ for $1 \le i \le n-1$, then b is **strictly** decreasing (i.e. b_1 gt b_2 gt ... gt b_{n-1}).

If there is no such array a, print a single integer -1.

Input

Each test contains multiple test cases. The first line contains the number of test cases t ($1 \le t \le 1000$). The description of the test cases follows.

The only line of each test case contains three integers x, y, n ($1 \le x$ lt $y \le 1000$,$3 \le n \le 1000$).

Output

For each test case, output n integers a_1,a_2,...,a_n. If there are multiple solutions, print any of them.

If there is no solution, print a single integer -1.

Example

Input
3 1 4 3 1 3 3 100 200 4

Output
1 3 4 -1 100 150 180 200

Note

In the first test case, a=[1,3,4], which is strictly increasing. Next, $b_1=a_2-a_1=3-1=2$, $b_2=a_3-a_2=4-3=1$, thus b=[2,1], which is strictly decreasing.

In the second test case, there is no array a that satisfies all the conditions above.